

A simulator for adaptive tutoring

Student models, knowledge tracing, question selection, and what we can and cannot measure with them

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1. What the simulator is for

A tutoring system has to choose the next question without being able to see what the student knows. To study that choice we need a student whose knowledge we *can* see, so that any claim about the tutor can be checked against the truth rather than against another estimate. Everything here is built around that one idea: the simulator owns a hidden state, the tutor never sees it, and every measurement compares the two.

The simulator has three replaceable parts, and keeping them separate is what makes the measurements interpretable.

Part	Sees the truth?	Question it answers
Student model	it <i>is</i> the truth	what does the learner actually know, and how does that change?
Knowledge tracing	no	what does the tutor believe, given only right and wrong answers?
Question selection	no	given that belief, what should be asked next?

Because they are separate, one can be varied with the others held fixed. That is the only way to attribute an effect: if the student and the tracer are fixed and only the scheduler changes, any difference is the scheduler's.

2. The curriculum

A curriculum is a directed acyclic graph of concepts. Each concept carries a tier (its depth), a list of prerequisites, and a bank of practice questions. Curricula are data, stored as JSON, so a new one is a new file rather than a code change.

The default graph is deliberately abstract and small: eight concepts named by tier, **A1**, **A2** at the top with no prerequisites, then **B1**, **B2**, **B3**, then **C1**, **C2**, and **D1** as the single capstone. A name states its own depth. The shape is a symmetric diamond, and two properties of it matter. The zone of proximal development starts at exactly two concepts, small enough to read at a glance; and **B2** is a prerequisite of both **C1** and **C2**, so a single weak concept visibly blocks two downstream. A 40-concept language curriculum is kept alongside it so that earlier results stay reproducible.

3. The student models

3.1 The plain BKT student

Each concept is a hidden coin: known, or not. Four parameters govern it.

	Meaning	Default
$p(L_0)$	probability the concept is known before any question	0.15
$p(T)$	probability an unknown concept becomes known at an attempt	0.12
$p(G)$	probability of a correct answer while it is unknown	0.25
$p(S)$	probability of a wrong answer while it is known	0.10

At each attempt on concept c , with $K_c \in \{0, 1\}$ the hidden state:

$$\Pr(\text{correct}) = \begin{cases} 1 - p(S), & K_c = 1, \\ p(G), & K_c = 0, \end{cases} \quad \text{then} \quad K_c \leftarrow 1 \quad \text{with probability } p(T) \text{ if } K_c = 0. \quad (1)$$

The student answers from the state it was in, and only then transitions. A learning event therefore shows up on the *next* question, not the one that caused it, which is why a student can be seen learning a concept on a question they got wrong. This ordering is not cosmetic: it is the ordering the tutor’s update assumes, and getting it backwards silently breaks calibration.

There is no forgetting, no question difficulty, and by default no effect of prerequisites on learning. That is the point. It is the simplest student for which the tutor’s arithmetic has a known right answer.

3.2 Two student models: S1 and S2

S1 Prerequisites do not affect learning. A concept is learned at $p(T) = 0.2$ whenever it is asked, whatever else the student knows.

S2 A concept is learned at $p(T) = 0.2$ once *all* of its prerequisites are truly known, and at $p(T) = 0.05$ otherwise.

S2 is the smallest change that makes the order of questions matter. Under S1 a question is worth the same whenever it is asked, so no scheduler can beat any other by more than noise; under S2 a question asked too early is worth a quarter as much.

3.3 The continuous student

The original student is also still available: mastery is a real number in $[0, 1]$, questions carry difficulty, prerequisites gate the learning rate continuously, and knowledge decays with elapsed time. It is the right model for studying forgetting and spaced practice, and the wrong one for checking the tutor’s arithmetic, because it has no binary truth to compare a probability against.

4. Knowledge tracing

The tutor keeps one number per concept: b_c , its belief that the student knows c . It sees only right and wrong answers. After an observation X it applies Bayes' rule and then the learning transition:

$$b^+ = \begin{cases} \frac{b(1-p(S))}{b(1-p(S)) + (1-b)p(G)}, & X = \text{correct}, \\ \frac{bp(S)}{bp(S) + (1-b)(1-p(G))}, & X = \text{wrong}, \end{cases} \quad b_c \leftarrow b^+ + (1-b^+)p(T). \quad (2)$$

Timing. The belief written by this update is the belief about the state the student is in *going into the next question*, because the transition has been applied. The student's hidden state after its own transition refers to the same instant. Any comparison of belief against truth must use that pairing; reading the state before the transition instead shifts everything by one step and makes a correct tracer look badly calibrated. This is the single easiest way to produce a convincing but wrong result, so it is stated here and enforced in the tests.

Because the tutor's parameters can be set independently of the student's, the tutor can be made deliberately wrong, which is what Section 7 exploits.

5. Question selection

The zone of proximal development is the set of concepts the tutor believes are worth teaching now:

$$\text{ZPD}(t) = \{ c : b_c < \theta \wedge b_p \geq \theta \ \forall p \in \text{prereq}(c) \}, \quad (3)$$

for a mastery threshold θ . Note that it is defined entirely on *beliefs*: the tutor has no access to the student's real state.

Two rules are compared.

Q2, `uniform_zpd` uniform over the ZPD. A concept is never asked until its prerequisites are believed mastered.

Q1, `uniform_unmastered` uniform over every concept not yet believed mastered, ready or not.

Both are deliberately free of any bandit. The comparison is about prerequisite gating alone.

6. What is measured, and against what

Every metric reads the student's hidden state, never the tutor's belief.

Calibration. Bin every belief the tutor has ever held; within a bin, what fraction of the time was the student truly in the known state? A correct tracer lies on the diagonal.

Detection delay. We know the exact step at which each concept was learned. How many further questions *on that concept* pass before the tutor declares it mastered?

False alarm rate. Of all the declarations the tutor made, what share came before the student had actually learned the concept?

Concepts truly known. How many concepts the student really knows at the end. This is the only metric that can compare schedulers, because it owes nothing to what any tutor believes.

Every comparison is paired: the same seeds are used in every arm, so learner i is the same simulated person throughout and differences are read within-learner rather than between groups.

7. What the simulator has been used to establish

The tracer is correct. On the plain BKT student with a correctly specified tutor, mean belief is 0.61 against a true known-fraction of 0.62 over 40,000 observations; at four times the sample every bin sits within 0.03 of the diagonal. Sweeping each parameter in turn over 19 settings leaves the largest calibration gap at 0.024, so calibration is not an artefact of one parameter choice.

The two checks agree with each other. If beliefs are calibrated, the false-alarm rate at threshold θ must equal one minus the mean belief at the moment of declaration. Measured and predicted agree at every threshold. Two quantities computed by different routes landing on the same number is stronger evidence than either alone.

The scheduler only matters if the student model says it can. Crossing S1 and S2 with Q1 and Q2, 60 learners, 40 questions:

	Q1	Q2	Q2 – Q1	95% CI
S1	6.22	5.83	–0.38	± 0.31
S2	4.47	5.65	+1.18	± 0.50
interaction (S2 – S1)			+1.57	± 0.40

Concepts truly known, out of eight. The *sign* of the scheduler effect flips with the student model, and the interaction is well clear of zero. Because seed i is the same simulated person in all four cells, that interval is the paired one: the two per-student differences are correlated ($r = +0.59$), so treating them as independent would overstate the uncertainty. Against S2 the prerequisite-respecting rule is worth about one extra concept; against S1 it is mildly harmful, because when prerequisites do not gate learning the ZPD restriction is pure overhead and Q1 gets to work on all eight concepts at once. The prediction going in was that the scheduler would be invisible under S1; it is in fact slightly counterproductive, which is a sharper result and the same lesson.

Budget decides whether any of this is visible. At 120 questions on eight concepts every cell reaches 7.7 of 8 and no scheduler can differ from another. A comparison of question-selection rules is only meaningful below saturation, and the saturation point has to be reported alongside any result.

Misspecification degrades gracefully but not harmlessly. When the tutor’s assumed $p(G)$ is swept while the student stays fixed, the calibration gap is +0.004 at the truth and reaches –0.128 when the tutor underestimates guessing by 0.20; false alarms move from 6% to 30% across that range. With a population of students whose parameters are spread uniformly by ± 0.15 and a single fixed tutor, the gap is –0.037 and false alarms roughly double, from 6.2% to 11.7%. A single tutor model stays usable across a varied population, at a measurable cost.

8. Limitations

The plain BKT student has no forgetting, so nothing here speaks to retention or spaced practice; the continuous student exists for that and is not used in any of the checks above. All results are on one curriculum of eight concepts. Populations are 30 to 60 learners, enough for the reported intervals and not for fine distinctions. The tutor’s $p(T)$ against S2 is held constant even though

the student's varies with its prerequisites, so the tutor is mildly misspecified there by construction; whether it should instead gate $p(T)$ on its own belief about the prerequisites is open. And every number is a statement about this model, not about people: no result here has been checked against a real learner.

Reproducing. `python3 make_sanity.py` regenerates every figure and table quoted above into `sanity_outputs/`; `python3 test_sanity.py` and `python3 test_simulator.py` run 29 and 55 checks respectively, including the exactness checks that pin the update ordering (with $p(G) = p(S) = 0$ a correct answer must come from a known concept; adding $p(T) = 1$ forces the tutor's belief to equal the truth at every step). Seeds are fixed throughout.